JAY KRISHNA

AI/ML Engineer

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PROFESSIONAL SUMMARY:

- Around 8 years of experience in **AI, ML**, and **Data Engineering**, with a strong focus on designing and deploying scalable, production-grade solutions across cloud and enterprise environments.
- Skilled in **Generative AI**, including prompt engineering, **LLM fine-tuning**, **Retrieval-Augmented Generation (RAG)**, embeddings, and building LLM-integrated applications using **LangChain** and **LangGraph**.
- Expertise in end-to-end AI system architecture, model lifecycle management, and building production-ready **ML/DL** pipelines for both real-time and batch inference systems.
- Full ownership of the ML lifecycle: data ingestion, preprocessing, feature engineering, model development, hyperparameter tuning, validation, deployment, monitoring, and retraining.
- Strong foundation in classical and deep learning models including **Random Forest, XGBoost, LightGBM**, **SVM**, Logistic Regression, CNNs, RNNs, **LSTMs**, Isolation Forest, Autoencoders, **BERT**, **GPT**, and T5.
- Developed and deployed AI microservices using Flask and **FastAPI**, containerized with **Docker**, and orchestrated with **Kubernetes (AKS/EKS)** for high-availability production serving.
- Delivered end-to-end AI/ML workloads on **AWS** (**SageMaker, Lambda, S3, EC2, EMR**) and **Azure** (Azure ML, **Databricks**, Synapse, Data Factory, Azure OpenAI Service).
- Integrated AI with real-time streaming systems using **Apache Kafka**, **Spark Streaming**, and serverless functions across finance, retail, and healthcare domains.
- Implemented **CI/CD** for ML using **MLflow,** DVC, **GitHub** Actions, **Jenkins**, and **Terraform** for reproducible, version-controlled model deployments.
- Proficient in **Python, SQL, Scala**, and PySpark with hands-on experience in **Scikit-learn, TensorFlow**, **PyTorch, HuggingFace Transformers**, NumPy, Pandas, NLTK, and spaCy.
- Experienced with vector databases (**FAISS, Pinecone**, Weaviate) for semantic search, similarity matching, and recommendation systems.
- Solid grasp of **MLOps** best practices: model versioning, experiment tracking, monitoring (Prometheus/Grafana), A/B testing, canary deployments, drift detection, and rollback strategies.
- Strong background in Big Data technologies including Apache Spark, Hadoop ecosystem (HDFS, **Hive**, HBase, **Sqoop**, **Oozie**), and large-scale distributed data processing.
- Built real-time dashboards and analytics tools using **Power BI**, **Tableau**, and **Plotly** for business stakeholders and operational teams.

SKILLS:

Languages & Scripting	**Python**, **SQL**, Scala, **PySpark**, Bash, Shell Scripting
ML / DL Frameworks	Scikit-learn, **XGBoost**, **LightGBM**, TensorFlow, **PyTorch**, Keras, **Spark MLlib**, HuggingFace Transformers
Generative AI & LLMs	**GPT-4, BERT, T5, RAG**, Prompt Engineering, LLM Fine-tuning, **LangChain**, **LangGraph**, FAISS, Pinecone
NLP & Text Analytics	spaCy, NLTK, **SentenceTransformers**, Text Classification, Named Entity Recognition, Sentiment Analysis
Model Deployment	Flask, **FastAPI**, Docker, Kubernetes (AKS/EKS), ONNX, **TensorRT**, **REST APIs**
MLOps & CI/CD	**MLflow**, DVC, GitHub Actions, **Jenkins**, **Terraform**, Azure DevOps, **Model Versioning**, Drift Detection

Cloud - Azure	Azure ML, **Databricks**, Data Factory, **Synapse**, ADLS Gen2, **AKS**, Azure **OpenAI Service**, Azure AI Search
Cloud - AWS	**SageMaker**, S3, EC2, **Lambda**, EMR, **Redshift**, Glue, **DynamoDB**
Big Data & Streaming	**Apache Spark**, PySpark, **Hadoop** (HDFS, Hive, HBase, Sqoop, MapReduce), Kafka, **NIFI**, **Airflow**
Databases	**Oracle**, MySQL, **PostgreSQL**, **MongoDB**, **Cassandra**, Snowflake, DynamoDB
Monitoring & Visualization	**Prometheus**, **Grafana**, Power BI, **Tableau**, Plotly

WORK EXPERIENCE:

Client: Wells Fargo **OCT 2023 - PRESENT**

Role: AI/ML Engineer

Overview: The bank needed a smarter, faster way to catch fraudulent transactions in real time. The goal was to build an AI-powered fraud detection system that analyzes millions of transactions as they happen, spots suspicious patterns early, and alerts the fraud operations team before damage is done, all while keeping false alerts low and meeting strict banking compliance standards. Built on **Azure** using ML models, real-time streaming with Kafka, and a **GenAI**-powered investigation assistant using **RAG** and **LangChain** to help analysts review flagged cases faster.

Key Responsibilities:

- Involved in all phases of **SDLC** including requirements gathering, modelling, analysis, architecture design, development, testing, and production deployment of AI/ML fraud detection solutions.
- Designed an end-to-end ML lifecycle covering data ingestion, **feature engineering**, model training, real-time inference, and monitoring using **PySpark** pipelines in **Azure Databricks**, orchestrated through Azure Data Factory.
- Developed and deployed **Spark** applications using PySpark and **Spark-SQL** for large-scale transaction data extraction, transformation, and aggregation from multiple banking sources to identify suspicious activity patterns.
- Built fraud classification models using **XGBoost** and **Random Forest** trained on transaction amount, merchant category, **geolocation**, device fingerprint, and time-of-day features to detect fraudulent transactions with high precision and recall.
- Implemented real-time anomaly detection using **Isolation Forest** and **Autoencoder** models on streaming transaction data to flag unusual spending behavior and account takeover attempts.
- Developed time-series models using **LSTM** and **ARIMA** to detect seasonal fraud trends and predict high-risk transaction windows for proactive monitoring.
- Built a customer risk scoring engine using **Logistic Regression** and **Gradient Boosting** to assign dynamic risk scores to accounts based on historical behavior and transaction velocity.
- Developed an **NLP** pipeline using **GPT-4**, **BERT**, and LangChain to analyze fraud investigation notes, **Suspicious Activity Reports (SARs)**, and case narratives for pattern extraction and case prioritization.
- Built a RAG-based fraud investigation assistant using **FAISS** and **HuggingFace Transformers** (SentenceTransformers) to help fraud analysts quickly retrieve relevant historical cases, regulatory guidelines, and investigation playbooks.
- Orchestrated multi-step reasoning workflows using **LangGraph**, separating retrieval, reranking, and summarization stages to produce traceable, audit-ready investigation summaries.
- Containerized and deployed ML scoring services via **FastAPI**, **Docker**, and **Azure Kubernetes Service (AKS)** with autoscaling to handle high-throughput real-time fraud scoring with sub-second latency.
- Implemented model explainability using **SHAP** and **LIME** to provide interpretable fraud prediction reasons, meeting regulatory audit and compliance requirements under **BSA/AML** guidelines.

- Managed experiment tracking and model versioning using **Azure Machine Learning** integrated with **MLflow**, enabling versioned deployment, drift tracking, and rollback across dev/stage/prod environments.
- Developed real-time monitoring dashboards using **Prometheus** and **Grafana** to track fraud scoring latency, model accuracy, false positive rates, and data drift events.
- Created data pipelines in ADF using linked services to ingest and transform transaction data from Azure SQL, **Blob Storage**, and **Azure SQL** Data Warehouse for model training and reporting.
- Developed custom **Kafka** producers and consumers for real-time streaming ingestion of transaction events from core banking systems.
- Optimized large-scale transaction data processing using Spark optimization techniques such as **partition tuning**, **caching**, and **broadcast variables** to reduce pipeline execution latency.
- Migrated legacy on-premises fraud detection workflows into **MS Azure cloud**, designing end-to-end solutions covering data storage, integration, model serving, and **visualization**.

Technical Stack: Python, PySpark, Azure Databricks, ADLS Gen2, Azure Data Factory, Azure ML, Azure **OpenAI** Service, LangChain, LangGraph, RAG, FAISS, **XGBoost**, Random Forest, Logistic Regression, **Gradient Boosting**, LSTM, ARIMA, Isolation Forest, Autoencoder, SHAP, LIME, BERT, GPT-4, **HuggingFace Transformers**, FastAPI, **Docker**, AKS, MLflow, **Prometheus**, **Grafana**, Kafka, Spark-SQL, Hive, HBase, **Oracle**, **NIFI**

Client: Walmart **APR 2022 - MAY 2023**

Role: Machine Learning Engineer

Overview: Walmart needed better ways to predict product demand, understand customer buying behavior, and catch supply chain issues early. The goal was to build ML models that process massive amounts of retail transaction data, forecast what products will sell where and when, group customers by shopping patterns, and flag any unusual activity in the supply chain, helping the business save costs, reduce waste, and improve stock availability. Built on Azure **Databricks** with **Spark-based ML pipelines,** experiment tracking via **MLflow**, and Power BI dashboards for business reporting.

Key Responsibilities:

- Designed end-to-end machine learning pipelines covering data ingestion, feature engineering, model training, batch scoring, and monitoring using **PySpark** on **Azure Databricks**, orchestrated through **Apache Airflow.**
- Developed Spark jobs using **Scala** and **Python** on **YARN** for interactive and batch analysis, improving performance and optimization of existing data processing algorithms.
- Built demand forecasting models using **XGBoost**, LightGBM, and **ARIMA** to predict product demand patterns, reducing inventory costs and improving stock availability across regions.
- Implemented customer segmentation models using **K-Means** clustering on large-scale POS transaction data to identify purchasing behavior patterns and support targeted marketing strategies.
- Developed classification models using **Random Forest** and **Logistic Regression** for supply chain anomaly detection, flagging irregular shipment patterns and delivery exceptions.
- Engineered features from retail transaction data including rolling aggregates, seasonal indicators, and merchant recurrence patterns using Spark window functions and **PySpark** transformations.
- Used **Python** libraries (NumPy, SciPy, Pandas, NLTK, Matplotlib) for **ETL** processing and **NLP**-based analysis of product reviews and customer feedback data.
- Applied **SHAP** explainability techniques to provide interpretable feature importance insights, enabling business stakeholders to understand key drivers behind demand and segmentation predictions.
- Managed experiment tracking and model versioning using **MLflow** within Databricks, ensuring consistent model performance across evolving retail datasets.
- Deployed containerized **ML** scoring services using Docker and orchestrated **CI/CD** pipelines using **Jenkins** and **Git** for reproducible, version-controlled deployments.
- Implemented Spark SQL for query optimization based on business requirements and migrated legacy **MapReduce** programs into Spark **RDD** transformations for better performance.

- Recreated existing application logic in **Azure Data Lake**, Data Factory, and **Azure SQL Database** for cloud-native data processing and analytics.
- Implemented real-time streaming ingestion using **Kafka** and **Spark** Streaming for live **POS** data analytics and near real-time reporting.
- Developed visualizations and dashboards using **Power BI** for operational analytics, **KPI tracking**, and executive reporting.
- Successfully migrated data pipeline orchestration jobs from **Oozie** to **Airflow** for improved scheduling, monitoring, and maintainability.

Technical Stack: Python, PySpark, Scala, Azure Databricks, Azure Data Lake, Azure Data Factory, Azure SQL, **XGBoost**, LightGBM, **ARIMA**, Random Forest, Logistic Regression, K-Means, **SHAP**, MLflow, **Docker**, **Jenkins**, Git, Spark-SQL, Spark Streaming, **YARN**, Kafka, **Airflow**, **Oozie**, **NLTK**, NumPy, Pandas, Power BI

Client: UNITED HEALTH GROUP **MAR 2020 - APR 2022**

Role: Big Data Engineer

Overview: Worked on building and managing large-scale data pipelines on **Azure** to move, transform, and process healthcare data from multiple source systems. The focus was on designing reliable **ETL** workflows using Azure Data Factory and **Databricks** to support downstream analytics, reporting, and decision -making across the organization.

Key Responsibilities:

- Used **Azure Data Factory** extensively for ingesting data from disparate source systems and as an orchestration tool for integrating data from upstream to downstream systems.
- Automated jobs using different **triggers** (Event, Scheduled, and Tumbling Window) in ADF for reliable, scheduled data processing.
- Used **Cosmos DB** for storing catalog data and for event sourcing in order processing pipelines.
- Analyzed the data flow from different sources to target and provided the corresponding design architecture in Azure environment.
- Created high-level technical design documents and application design documents as per the business requirements.
- Created Build and Release definitions for **Continuous** Integration and Continuous Deployment using Azure **DevOps**.
- Created pipelines, **data flows**, and complex data transformations using ADF and PySpark with Databricks.
- Ingested data in mini-batches and performed **RDD** transformations using **Spark Streaming** for streaming analytics in **Databricks**.
- Created and provisioned different **Databricks** clusters for batch and continuous streaming data processing with required **libraries**.
- Integrated Azure Active Directory authentication to **Cosmos DB** requests and demonstrated features to stakeholders.
- Created numerous pipelines in Azure Data Factory v2 with different activities including Copy, Filter, **ForEach**, and Databricks notebooks.
- Created several Databricks Spark jobs with **PySpark** to perform table-to-table operations and data transformations.
- Worked with complex **SQL**, Stored Procedures, **Triggers**, and packages across large databases from various servers.

Technical Stack: Hadoop, Hive, **Oozie**, Spark, Spark SQL, **Python**, PySpark, Azure Data Factory, Azure SQL, Azure Databricks, Azure DW, **BLOB Storage**, Scala, **AWS**, **Linux**, Maven, **Oracle**, Zookeeper, MySQL, Snowflake

Client: EPAM **JAN 2018 - FEB 2020**

Role: Data Engineer

Overview: Started career building data pipelines and working with the **Hadoop** ecosystem to process, transform, and manage large volumes of data from multiple sources. Focused on ETL development, data validation, and supporting analytics teams with clean, reliable datasets using tools like Hive, **Sqoop**, **MapReduce**, and **AWS**.

Key Responsibilities:

- Administered, maintained, provisioned, patched, and managed **Cloudera** Hadoop clusters on **Linux** environments.
- Worked on analyzing **Hadoop** stack and different big data analytic tools including Hive, **HBase** database, and **Sqoop**.
- Utilized Python matplotlib and **Scikit-Learn** modules for basic prototype visualization and early machine learning experiments.
- Written multiple **MapReduce** programs for data extraction, transformation, and aggregation from 20+ sources with multiple file formats (**XML, JSON, CSV**).
- Worked in **AWS** environment for development and deployment of custom Hadoop applications on **EC2 and S3**.
- Created Oozie workflows for orchestrating Hadoop-based jobs including Sqoop imports and Hive scripts.
- Involved in file movements between **HDFS** and AWS **S3** for cross-platform data integration.
- Created Hive external tables, loaded data into tables, and performed queries using **HQL** for ad-hoc analytics.
- Performed data validation on ingested data using **MapReduce** by building custom models to filter invalid
- records and cleanse datasets.
- Handled importing of data from various sources, performed transformations using Hive and MapReduce, loaded data into **HDFS**, and extracted data from **MySQL** using **Sqoop**.
- Transferred data using **Informatica** tool from AWS S3 to AWS **Redshift** for data warehouse reporting.
- Involved in **ETL** processes using **SSIS** and generated reports using **SSRS** for business stakeholders.
- Managed cluster coordination services through Zookeeper, and installed, configured, and wrote Hive UDFs.

Technical Stack: Hadoop, Hive, HBase, Sqoop, **Python**, **Matplotlib**, Scikit-Learn, **Tableau**, HDFS, MapReduce, AWS, S3, **Redshift**, MySQL, **OLAP**, **Oozie**, HQL, SSIS, **SSRS**, MS SQL Server, HiveQL

CERTIFICATIONS:

- AWS Certified Machine Learning Specialty
- Microsoft Certified: Azure Data Engineer Associate
- Databricks Certified Generative AI Engineer Associate

EDUCATION:

- **Master's in Data Science**, *Saint Peter's University*, New Jersey, USA
- **Bachelor's in Computer Science**, *JNTUH*, Hyd, India